



PRODUCTION VERIFIED

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Geospatial Thermal Intelligence & AI Decision Platform

Deployment: flarex-gamma.vercel.app

1. Executive Vision & Problem Statement

Standard NASA FIRMS implementations only visualize where heat anomalies are detected. However, FIRMS data alone cannot answer the critical operational question: **"Is this heat an uncontrolled industrial disaster, a routine operational gas flare, a wildfire, or seasonal agricultural burning?"**

The FlameX Paradigm Shift

FlameX transforms raw satellite heat coordinates into contextual, explainable intelligence by fusing **satellite radiance (NASA FIRMS)**, **infrastructure geofences (OpenStreetMap)**, **10m land cover (ESA WorldCover)**, and a **30-day temporal baseline engine**.

2. The 5 Core Data Streams

Data Stream	Source & Resolution	Operational Role in FlameX
1. Thermal Radiance	NASA FIRMS (VIIRS 375m / MODIS 1km)	Ingests Fire Radiative Power (FRP in MW), brightness temperatures (T4/T5), detection confidence, and satellite timestamps.
2. Industrial GIS Context	OpenStreetMap (OSM) Vectors	Computes spatial proximity (distance in meters) to refineries, chemical SEZs, power plants, steel mills, and mines.
3. Land-Cover Context	ESA WorldCover (10m Pixel Grid)	Categorizes surface landuse: <i>Industrial/Built-up, Dense Forest, Cropland, Mining/Bare Soil, Water.</i>
4. Optical Verification	Sentinel-2 MSI (10m Bands)	Provides true-color (B04, B03, B02) and SWIR (B12) visual verification of smoke plumes and structural damage.
5. Population Exposure	Demographic Population Grids	Measures distance to nearest residential settlement zones to escalate hazard severity to CRITICAL.

3. How the AI Predicts: The Detective Methodology

When a heat anomaly is detected, FlameX executes a multi-feature decision algorithm answering three fundamental questions:

1. WHERE is the Heat?

Checks underlying terrain via ESA WorldCover and distance to industrial assets via OSM. Anomaly within 500m of a refinery is tagged as industrial.

2. WHAT is Normal for this Spot?

Queries the 30-day baseline engine. If a refinery flare burns at 108 MW every day (28/30 days), it is normal. If it surges to 380 MW (3.6× baseline) with no recurrence, it is an emergency fire.

3. WHO is at Risk?

Evaluates human settlement proximity. If high radiance is within 1,000m of a populated town, FlameX escalates the incident to **CRITICAL DISPATCH**.

4. Case Study: Normal Gas Flare vs Uncontrolled Fire

The table below illustrates how the AI differentiates a routine refinery flare stack from a severe chemical plant fire:

Evaluation Parameter	Scenario A: Jamnagar Mega Refinery	Scenario B: Dahej Petrochemical SEZ
Thermal Power (FRP)	112.0 MW	380.0 MW (Catastrophic Spike)
30-Day Historical Baseline	108.0 MW	105.0 MW
Baseline Multiplier (FRP / Baseline)	1.04× (Normal Variance)	3.6× ABOVE BASELINE
30-Day Recurrence Frequency	28 / 30 days (Daily continuous flaring)	3 / 30 days (Sudden unexpected onset)
Land Cover (ESA WorldCover)	Industrial / Built-up	Industrial / Built-up
Facility Proximity (OSM)	45 m from Refinery Flare Unit A	65 m from Chemical Refinery Tank Farm
Population Buffer	3.8 km (Safe industrial buffer)	450 m from Dahej Coastal Settlement
AI Classification Verdict	🟢 GAS FLARE (NORMAL)	🔴 INDUSTRIAL FIRE (CRITICAL)

5. Multi-Source Explainability Checklist ("Why?")

Rather than acting as an opaque black box, FlameX synthesizes verified proof points for disaster response teams and regulatory judges:

AI Evidence Synthesis for Incident #FL-102 (Dahej SEZ)

- ✓ 65 m from XYZ Petrochemical Refinery Tank Farm (OSM infrastructure match)
- ✓ Industrial / Built-up land cover (ESA WorldCover 10m high-resolution classification)
- ✓ Extreme thermal radiance: 380.0 MW with skin temperature of 139°C (NASA FIRMS VIIRS)
- ✓ 3.6× above 30-day historical baseline (105.0 MW typical operating level)
- ✓ Low temporal recurrence (3/30 days) indicating sudden catastrophic onset rather than routine flaring
- ✓ Population vulnerability: 450 m from Dahej residential settlement
- ✓ Zero adjacent dense forest or agricultural burning footprint (Eliminates natural wildfire/stubble)

6. Technical Stack & Architecture Specification

Layer	Technology	Architecture Details
Frontend Platform	Next.js 16 (App Router + React 19)	Dynamic client components with fast Turbopack compilation. Zero hardcoded stats.
2D Geospatial Engine	MapLibre GL JS (WebGL)	High-resolution satellite basemaps with live multi-ring danger pulses and GIS boundaries.
3D Globe Overview	Three.js & React Three Fiber	Cinematic Earth sphere with orbital particle shaders and camera panning.
Charts & Curves	Recharts	Real-time SVG 7-day radiative trajectory curve with dashed baseline reference lines.
Database & Engine	SQLite (better-sqlite3)	Write-Ahead Logging (WAL), automatic /tmp serverless mount for Vercel deployment.
AI Engine	FlameX Spatial Ensemble v1.2	6-class probability distribution, abnormality scoring, and natural language copilot.
Cloud Deployment	Vercel Production Edge	Live on <code>https://flarex-gamma.vercel.app</code> with GitHub CI/CD integration.